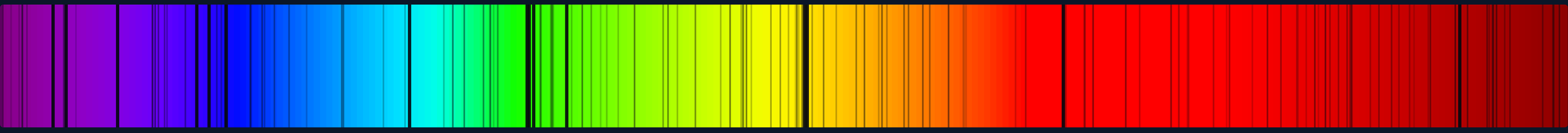
How to find a planet you cannot see

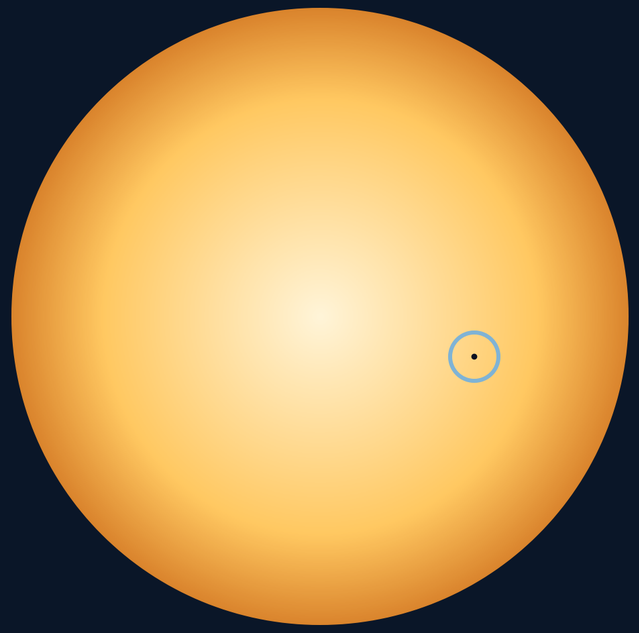
Two tiny signals: the shadow and the wobble

presentation-kit · expressive example · 2026



THE SHADOW

When a planet passes in front, its star dims a little



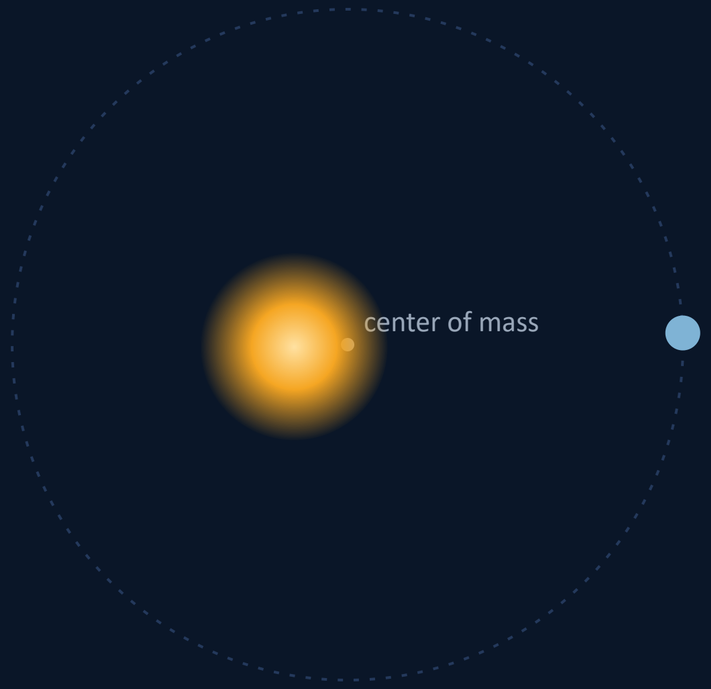
an Earth-sized planet (circled)

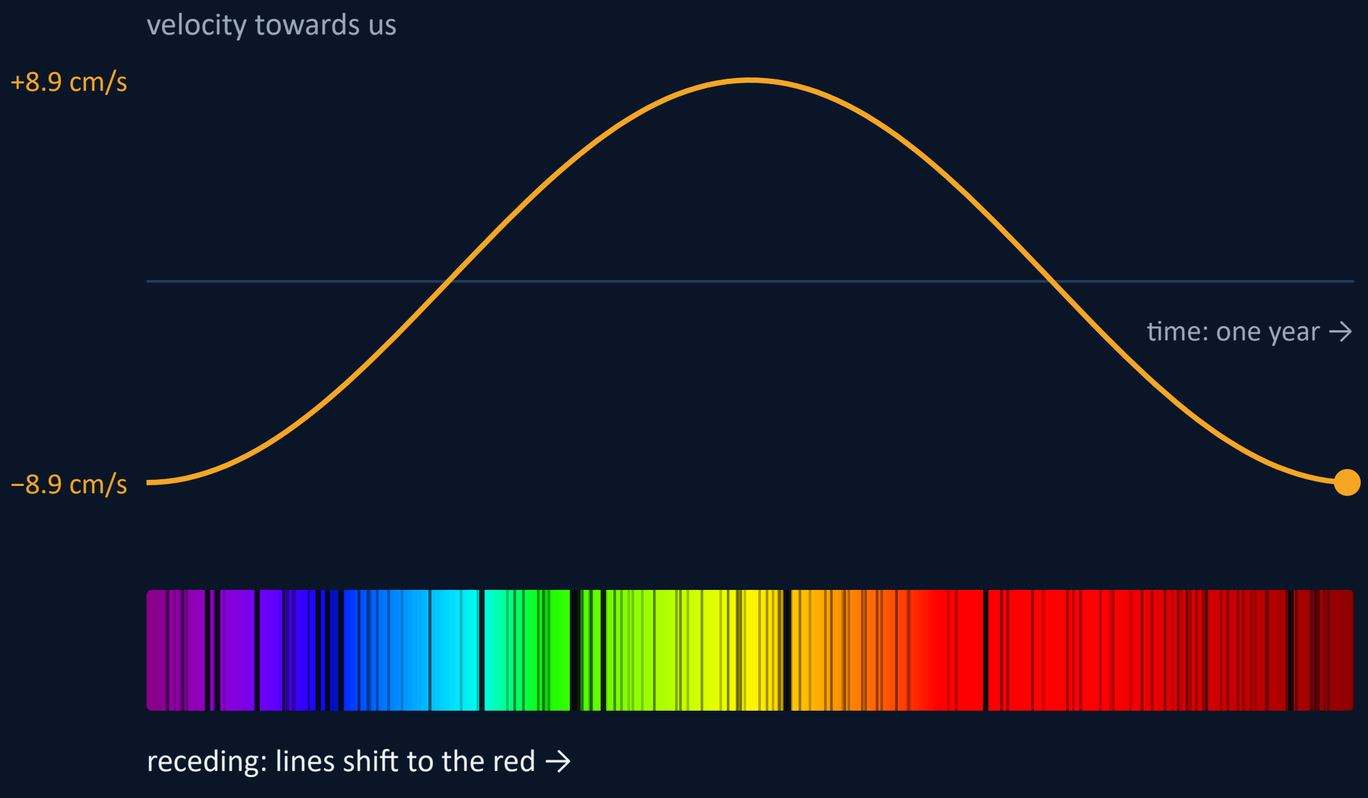


Source: Depth = (R_planet / R_star)², with IAU (2015) nominal radii: Jupiter in front of the Sun, 1.06 %; Earth, 84 ppm. No limb darkening; Earth's size not to scale in the curve.

THE WOBBLE

The star wobbles too: Earth moves the Sun by only 9 cm/s

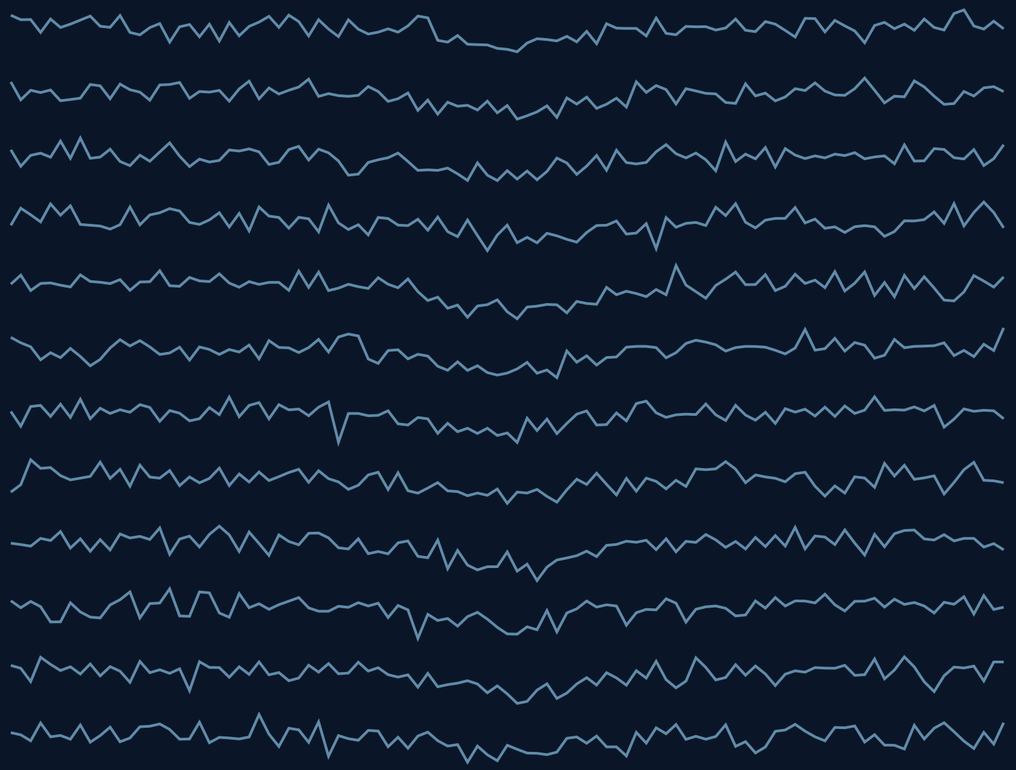




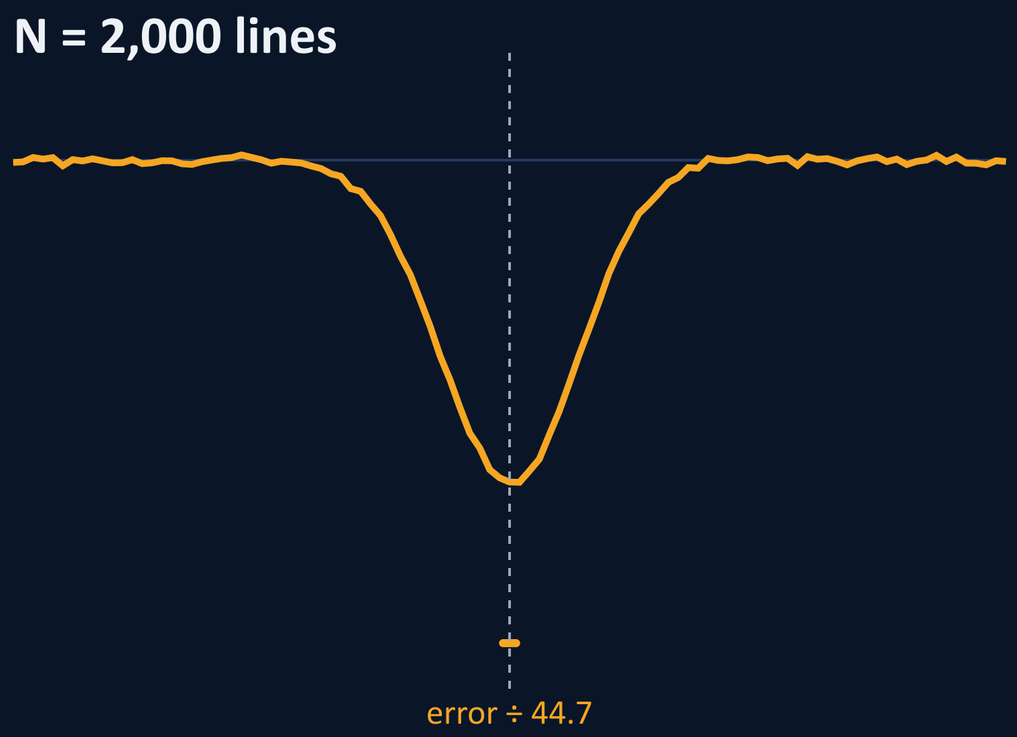
Source: Kepler's third law, circular orbit: Earth induces 8.9 cm/s on the Sun; Jupiter, 12.5 m/s. Motion and shift exaggerated so they can be seen.

THE TRICK

A single line is noise; adding thousands, the error drops as $1/\sqrt{N}$



… and 1,988 more lines



Source: Simulation with identical Gaussian lines and photon noise; these are not real lines. The velocity error drops as 1/√N: with 2000 lines, about 45 times smaller.

WHAT IS MEASURED

Together, the two signals tell what the planet is made of

- **The shadow gives the size:** the depth is $(R_{planet} / R_{star})^2$.
- **The wobble gives the mass** (the minimum: it depends on the orbit's inclination).
- **Size and mass give the density:** rock like Earth or gas like Jupiter.

Source: Standard relations of the transit and radial-velocity methods.

THE SCALE

Finding another Earth means signals more than a hundred times smaller than a Jupiter's

how much the star wobbles (velocity)



how much it dims in transit (depth)



Source: Own computation: velocity (Kepler) 12.5 m/s vs 8.9 cm/s, ÷139; depth (IAU 2015) 1.06 % vs 84 ppm, ÷126. Logarithmic scales.

HOW IT WAS MADE

This talk is a text script, five scenes and a profile

- **script.md** (and script.es.md): one conclusion-title per slide, with its source and notes.
- **animations.js:** five scenes built with `K.scene`, one state per step.
- **night profile:** colors and fonts; `kit.py build` produces the HTML and the .pptx.

Source: presentation-kit: engine/kit.py, engine/anim.js, profiles/night.

APPENDIX

Sources

- Nominal radii: IAU 2015 Resolution B3 (Sun 695,700 km; Earth equatorial 6,378.1 km; Jupiter equatorial 71,492 km).
- Masses: Sun 1.989×10^{30} kg; Earth 5.972×10^{24} kg; Jupiter 1.898×10^{27} kg. G = 6.674×10^{-11} m^3 kg^{-1} s^{-2}.
- Induced velocity: Kepler's third law for a circular orbit.
- Everything is computed in animations.js; the animations exaggerate motion so it can be seen.